MAcRIO

Problem Statement

Big multinational food corporations frequently use a double standard when selling everyday food items across Asian, African, and Latin American markets compared to the US, Canada, and the EU. For the exact same brand and product, companies swap out quality ingredients for cheap, degraded alternatives like low-grade palm oil, excess refined sugars, artificial dyes, and banned preservatives.

At the same time, current calorie and macro tracking apps are built around Western diets and completely miss regional groceries, local street foods, and local languages in priority markets like Singapore, Malaysia, India, and Indonesia. On top of that, most people do not have an easy way to check if their daily cooking utensils, scratched non-stick pans, cheap aluminum pots, and plastic containers are leaching toxins and microplastics into their food.

What MAcRIO Does

- **Cross-Border Ingredient & Barcode Scanner:** Users scan barcodes or snap a photo of ingredient labels in Singapore, Malaysia, India, and Indonesia. MAcRIO compares the local product directly against its US, Canadian, and EU versions to expose where big brands used cheaper, degraded ingredients or unsafe additives.
- **Local Meal Planning & Food Avoidance:** Users take photos of local meals and street food to get accurate macro breakdowns. The AI meal planner tells users which specific foods and ingredients to avoid based on their personal health goals, explained clearly in local languages (Malay, Indonesian, Hindi, Tamil, and Mandarin).
- **Plastic & Utensil Safety Check:** Users share photos of their kitchen utensils, pans, and plastic food containers. The app checks plastic resin codes (1 to 7) and surface wear to flag bad plastics, scratched Teflon, and cheap aluminum that leach chemicals into hot meals.
- **All-in-One Health Guide:** Gives practical daily advice on healthier local brand swaps, better cooking habits, and hitting macro goals without forcing users to rely on Western apps that do not recognize their food.

How We Use IBM Z and LinuxONE

Native s390x Container Backend: Our Python backend (FastAPI) and PostgreSQL database run inside Docker containers built directly for the s390x architecture on IBM LinuxONE, allowing us to handle massive global barcode and ingredient tables without lag.

Real-Time AI with IBM Telum: When a user is standing in a grocery aisle scanning a barcode or taking a picture of a meal, speed matters. We use the IBM Telum processor and its on-chip AI acceleration to run fast OCR label reading and image detection in real time.

Private Health Profiles with IBM Hyper Protect: Because users log personal health conditions like diabetes, severe allergies, high blood pressure, and gut issues, we isolate user data inside IBM Hyper Protect Virtual Servers. Even as the AI plans their meals, their private medical info stays completely locked down.

Datasets Used

- **Open Food Facts (Global & Regional):** Used to match and compare product ingredients sold in ASEAN, India, Africa, and LATAM against US, Canadian, and EU formulas.
- **Regional Nutrition Tables:** Singapore Health Promotion Board (HPB), Malaysian Food Composition Database (MyFCD), Indian Food Composition Tables (IFCT), and Indonesian food tables for local macro tracking.
- **Food Safety Watchlists:** US FDA, EU EFSA, Singapore SFA, and India FSSAI databases to flag banned or restricted food dyes, preservatives, and cheap fillers.
- **Food & Kitchenware Image Sets:** Regional street food images for macro estimation alongside plastic code and worn cookware images for utensil safety detection.

Target Audience

MAcRIO is built first for everyday shoppers, families, fitness trackers, and people managing medical diets across Singapore, Malaysia, India, and Indonesia, before expanding across wider Asian, African, and Latin American markets. It gives everyday people a simple way to see what is really inside their food and kitchenware.